# Fourier Packets and the Continuum Dirac Operator

From finite spinor dynamics to a self adjoint Hamiltonian on the three torus

## From finite spinors to a continuum Hamiltonian

The preceding calculus becomes a concrete field construction when Fourier modes are used as packets. Each mode carries a four-component spinor and an explicit Hermitian matrix. The field is reconstructed through an isometric Fourier map. Exact norm preservation, completion, approximation, and control of the adjoint domain justify the passage from finite calculations to continuum analysis.

The setting is a free spinor on a fixed flat three-torus. For every real mass parameter, the resulting Hamiltonian is densely defined, self-adjoint, and closed. Its domain is exactly the first-order weighted Fourier square-summability domain. These assertions include the operator domain, not merely a formal differential expression.

The Hamiltonian and the modewise unitary evolution are first constructed separately. The following chapter, *The Dirac Group and Its Generator*, connects them by proving strong continuity, equality of the full generator domain with the Hamiltonian domain, and the spectral-exponential identity. It also develops graph-domain and strong-resolvent convergence for the centered Fourier family and a quadratic spacing estimate under third-order Fourier regularity.

> **The role of this construction**
>
> Finite Clifford identities determine each mode. Hilbert-space completion reconstructs the field. Adjoint-domain equality determines the self-adjoint operator. The generator argument then identifies its evolution. These are distinct steps in one continuous mathematical construction.

The operator construction, full generator-domain identification, centered graph and strong-resolvent limits, and third-order moment estimates have recorded Lean verification. The abstract spectral-functional-calculus identity remains a written deduction without a separate formal API linkage. The technical appendices identify the exact formal statements and evidence. None of these mathematical results by itself identifies an observed particle or supplies a dynamical spacetime geometry.

## Reading routes and evidence

Readers seeking the construction can begin with the conventions and Clifford matrices, follow the finite-packet estimates, and continue through coefficient completion to the adjoint calculation. Readers already comfortable with free Dirac matrices can begin at the full coefficient Hilbert space. The exact three-mode example gives a concrete state that can be followed through both continuum and centered evolution.

There are three kinds of statement in this volume. A *checked correspondence* restates mathematics represented by named declarations in the retained formal source. An *ordinary mathematical deduction* has its proof in the text but is not automatically assigned to a Lean declaration. An *illustration* is an explicitly bounded Python evaluation of a formula. The figure record specifies its parameters and numerical checks.

The finite-packet propositions retain the identifiers PH01 to PH12 from the earlier traditional paper. This preserves their correspondence with its worked example and source notes. The later sections use numbered theorems for the completed-space and unbounded-operator results. A correspondence annotation identifies the checked source layer without asserting that the prose proof is a literal transcription of a tactic script.

| Object | Meaning in this volume |
|---|---|
| Mode Hamiltonian | An everywhere-defined Hermitian matrix on a four-dimensional spinor space. |
| Coefficient evolution | A modewise unitary map on the complete coefficient Hilbert space. |
| Physical evolution | The conjugate of that map under the checked Fourier equivalence. |
| Continuum Hamiltonian | An unbounded partial linear operator with its domain carried explicitly. |
| Centered approximation | The multiplier with kinetic symbol $\sin(hp_j)/h$ on the same integer mode index set. |
| Generator identification | The theorem in the following chapter connecting the full-domain derivative and spectral exponential to the constructed Hamiltonian. |

The centered approximation is defined on continuum torus functions. A spatially sampled finite grid would require its own sampling, aliasing, and embedding conventions. Similarly, first-order regularity is expressed here by the precise Fourier condition that the sources prove. An identification with every possible weak-derivative or Sobolev-library convention would require a further equivalence theorem.

No finite count of successful numerical evaluations replaces the universal statements in the manuscript or their formal counterparts. Conversely, checking a mathematical model does not establish that the model describes observed matter. The physical setting here is a free spinor on a fixed flat periodic space, in natural units.

# Resolution and finite dynamics

## 1 Packets and transfer blocks

The useful starting point is a resolved operator rather than a preferred coordinate matrix. Let $V$ be a complex vector space with a finite family of projectors $(P_a)_{a\in A}$ satisfying

$$P_aP_b = \delta_{ab}P_a, \qquad \sum_{a\in A} P_a = I.$$

The component $P_a v$ is the packet of $v$ selected by $a$. For a linear map $L : V \to V$, its transfer block from packet $a$ to packet $b$ is

$$L_{ba} = P_bLP_a.$$

Completeness gives $L = \sum_{a,b} L_{ba}$. Both indices matter. Keeping only $P_aLP_a$ would remove every transfer between distinct packets.

For two operators $L$ and $M$, insert the resolution between them:

$$P_cMLP_a = \sum_{b\in A}(P_cMP_b)(P_bLP_a).$$

This is block multiplication written without a choice of basis inside each packet. The intermediate index records the interface through which a composition passes. It is also the reason that a change of resolution must specify how fine packets assemble into coarse packets.

Suppose $P_A = \sum_{a\in C(A)} P_a$ and $P_B = \sum_{b\in C(B)} P_b$. Then

$$P_BLP_A = \sum_{b\in C(B)} \sum_{a\in C(A)} P_bLP_a.$$

The identity follows by distributing two finite sums. It says exactly which fine transfers a coarse observation retains. A scalar summary or a diagonal-only summary may retain less information and therefore needs its own loss statement.

These finite identities provide the notation used throughout the wider collection. The free Fourier model has an additional property: translation invariance makes its Hamiltonian diagonal in the mode label. Each diagonal block is still a nontrivial four-spinor matrix. The simplification belongs to this model, not to every operator that can be expressed in packets.

### 1.1 What changes at completion

The finite identity $\sum_a P_a = I$ becomes an approximation statement for an infinite orthogonal family. Finite partial sums must converge in the chosen topology. On the coefficient Hilbert space, canonical finite truncations converge in norm. For an unbounded Hamiltonian, approximating the state alone is insufficient to control its image. The graph approximation later in the volume controls both quantities at once.

This distinction explains the structure of the proof. Exact finite algebra supplies local formulas. Hilbert stability controls their extension to all square-summable fields. Weighted regularity specifies which fields support an unbounded operator action. Each step carries a different mathematical obligation.

## 2 Conventions and finite packet space

We use natural units $\hbar = c = 1$. Space is the flat unit torus

$$\mathbb{T}^3 = (\mathbb{R}/\mathbb{Z})^3,$$

equipped with normalized Haar measure $dx$, so $\int_{\mathbb{T}^3} 1\, dx = 1$. Integration of a periodic representative may equivalently be taken over $[0,1)^3$ with Lebesgue measure. A Fourier mode is $n = (n_1, n_2, n_3) \in \mathbb{Z}^3$, its momentum is $p(n) = 2\pi n$, and its scalar plane wave is

$$e_n(x) = e^{2\pi i n\cdot x} = e^{ip(n)\cdot x}.$$

Components of a spinor are numbered $0, 1, 2, 3$, in the exact order of the source representation. For $v, w \in \mathbb{C}^4$,

$$\langle v, w\rangle = \sum_{r=0}^{3} \overline{v_r} w_r, \qquad \|v\|_2^2 = \sum_{r=0}^{3} |v_r|^2.$$

The inner product is conjugate-linear in its first argument and linear in its second. The adjoint is therefore the conjugate transpose. Unless indicated otherwise, a spinor norm is this Euclidean norm and an operator norm is its induced operator norm. The field pairing is

$$\langle f, g\rangle_{L^2} = \int_{\mathbb{T}^3} \langle f(x), g(x)\rangle\, dx$$

on $L^2(\mathbb{T}^3; \mathbb{C}^4)$. Fix a finite set $S \subset \mathbb{Z}^3$. A packet with amplitudes $a_n \in \mathbb{C}^4$ is

$$f_a(x) = \sum_{n\in S} e_n(x) a_n.$$

The associated coefficient space is the finite Hilbert direct sum

$$\mathcal{K}_S = \bigoplus_{n\in S} \mathbb{C}^4, \qquad \|a\|_{\mathcal{K}_S}^2 = \sum_{n\in S} \|a_n\|_2^2.$$

The packet space $\mathcal{F}_S = \{f_a : a \in \mathcal{K}_S\}$ will be identified isometrically with $\mathcal{K}_S$. Its elements are smooth periodic functions, so the differential and centered-translation calculations below raise no regularity question. Empty support is allowed: both spaces then contain only zero and all packet identities reduce to $0 = 0$.

The parameters $m, h, t$ are arbitrary real numbers. The mathematics does not require $m \geq 0$; the usual physical rest-mass interpretation uses that additional convention. The shell value $E$ is always a nonnegative square root, even when $m < 0$. Mesh limits in the checked statements are $h \to 0^+$. Some explicit estimates proved here hold for every real $h$, since the centered symbol is even in $h$.

## 3 The coordinate Clifford representation

The spatial generators and mass generator are

$$\alpha_x = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad \alpha_y = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix},$$

$$\alpha_z = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \qquad \beta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

In particular, $\alpha_y v = (-iv_3, iv_2, -iv_1, iv_0)$; this fixes a sign that cannot be changed silently in a worked example. The matrices act on column spinors. Grouping the first two and last two components gives their block form without any coordinate permutation:

$$\alpha_j = \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix}, \qquad \beta = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix},$$

where

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

**Proposition PH01 (Hermitian Clifford relations).** The four displayed matrices are Hermitian involutions. Distinct spatial generators anticommute, and each spatial generator anticommutes with $\beta$:

$$\alpha_j^* = \alpha_j, \quad \beta^* = \beta, \quad \alpha_j^2 = \beta^2 = I_4,$$

$$\alpha_j\alpha_k + \alpha_k\alpha_j = 0 \quad (j \neq k), \qquad \alpha_j\beta + \beta\alpha_j = 0.$$

*Proof.* The real diagonal entries of $\beta$ are $1, 1, -1, -1$, so its adjoint and square have the stated forms. Each $\sigma_j$ is Hermitian and squares to $I_2$, by direct multiplication. Their products are

$$\sigma_x\sigma_y = i\sigma_z, \qquad \sigma_y\sigma_z = i\sigma_x, \qquad \sigma_z\sigma_x = i\sigma_y,$$

with reversed products equal to their negatives. For example,

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i\sigma_z.$$

The other two follow by the same two-by-two multiplication; reversal follows by taking adjoints. The block form gives

$$\alpha_j^* = \begin{pmatrix} 0 & \sigma_j^* \\ \sigma_j^* & 0 \end{pmatrix} = \alpha_j, \qquad \alpha_j\alpha_k = \begin{pmatrix} \sigma_j\sigma_k & 0 \\ 0 & \sigma_j\sigma_k \end{pmatrix}.$$

Thus $\alpha_j^2 = I_4$ and the spatial anticommutators vanish. Finally,

$$\alpha_j\beta = \begin{pmatrix} 0 & -\sigma_j \\ \sigma_j & 0 \end{pmatrix}, \qquad \beta\alpha_j = \begin{pmatrix} 0 & \sigma_j \\ -\sigma_j & 0 \end{pmatrix},$$

whose sum is zero. Every identity is an identity of the exact source matrices. □

## 4 Continuum and centered mode Hamiltonians

For $p \in \mathbb{R}^3$, define

$$H(m,p) = m\beta + p_x\alpha_x + p_y\alpha_y + p_z\alpha_z.$$

Its coordinates are

$$H(m,p) = \begin{pmatrix} m & 0 & p_z & p_x - ip_y \\ 0 & m & p_x + ip_y & -p_z \\ p_z & p_x - ip_y & -m & 0 \\ p_x + ip_y & -p_z & 0 & -m \end{pmatrix}.$$

On mode $n$, the continuum Hamiltonian is $H_n = H(m, p(n))$. It is the symbol of the differential expression

$$-i\sum_{j=x,y,z} \alpha_j\partial_j + m\beta$$

on the finite packet space: $\partial_j e_n = ip_j(n)e_n$, so applying the expression to $e_n a_n$ gives $e_n H_n a_n$. For real spacing $h$, set

$$q_j(h,p) = \begin{cases} \sin(hp_j)/h, & h \neq 0, \\ p_j, & h = 0. \end{cases}$$

Write $q_h(n) = q(h, p(n))$ and $H_{h,n} = H(m, q_h(n))$. The function $q_j$ is even in $h$, and its value at zero is justified by $\sin z/z \to 1$. For $h \neq 0$, a centered translation difference on a periodic smooth function is

$$(D_{h,j}f)(x) = \frac{f(x + he_j) - f(x - he_j)}{2h}.$$

Here $e_j$ is the j-th spatial unit vector, distinct from the Fourier characters $e_n$ and spinor vectors. Its action on a plane wave is exactly

$$D_{h,j}e_n = e_n\frac{e^{ihp_j(n)} - e^{-ihp_j(n)}}{2h} = i\frac{\sin(hp_j(n))}{h}e_n.$$

The centered differential expression $-i\sum_j \alpha_j D_{h,j} + m\beta$ therefore has symbol $H_{h,n}$. Its signs follow from $-i \cdot i = 1$. At $h = 0$, the symbol equals $H_n$ by definition. These formulas take place on the continuum torus and use no sampled spatial data.

**Proposition PH02 (Hermiticity and the mass shell).** For every real $m$ and $p \in \mathbb{R}^3$,

$$H(m,p)^* = H(m,p), \qquad H(m,p)^2 = (m^2 + |p|^2)I_4.$$

In particular, $H_n, H_{h,n}$ are Hermitian for all real $m, h$, and

$$H_n^2 = E_n^2 I_4, \quad E_n = \sqrt{m^2 + |p(n)|^2},$$

$$H_{h,n}^2 = E_{h,n}^2 I_4, \quad E_{h,n} = \sqrt{m^2 + |q_h(n)|^2}.$$

*Proof.* Adjoint is additive, and a real multiple of a Hermitian matrix is Hermitian. PH01 therefore proves the first identity. Expanding the square gives the square terms $m^2\beta^2 + \sum_j p_j^2\alpha_j^2$, the terms $mp_j(\beta\alpha_j + \alpha_j\beta)$, and the terms $p_jp_k(\alpha_j\alpha_k + \alpha_k\alpha_j)$ for $j < k$. The anticommutators vanish and all four squares are $I_4$, giving $(m^2 + \sum_j p_j^2)I_4$. Substitution proves both mode statements. □

Hermiticity here is a statement about matrices acting on all of $\mathbb{C}^4$. Every linear map on this finite space is bounded and everywhere defined. This is not a construction of the adjoint or domain of an unbounded differential operator on the full field space.

## 5 Spectral projectors and exact mode evolution

Let $H = H(m,p)$ and $E = \sqrt{m^2 + |p|^2}$. Define the exact propagator

$$U_H(t) = e^{-itH} = \sum_{k=0}^{\infty} \frac{(-itH)^k}{k!}.$$

The series converges absolutely in operator norm, being dominated by $\sum_k (|t|\|H\|)^k/k!$. This is exact continuous-time evolution, with no time step or series truncation.

**Proposition PH03 (spectral splitting and closed propagator).** If $E > 0$, define

$$P_+ = \frac{1}{2}\left(I_4 + \frac{H}{E}\right), \qquad P_- = \frac{1}{2}\left(I_4 - \frac{H}{E}\right).$$

Then $P_\pm^* = P_\pm$, $P_\pm^2 = P_\pm$, $P_+P_- = P_-P_+ = 0$, $P_+ + P_- = I_4$, and $HP_\pm = \pm EP_\pm$. Both projectors have rank two, and

$$U_H(t) = e^{-itE}P_+ + e^{itE}P_- = \cos(tE)I_4 - i\frac{\sin(tE)}{E}H.$$

If $E = 0$, then $H = 0$ and $U_H(t) = I_4$. Equivalently, the closed expression is valid with the continuous convention $\sin(tE)/E = t$ at $E = 0$.

*Proof.* For $E > 0$, the Hermitian matrix $K = H/E$ satisfies $K^2 = I_4$. Thus

$$P_\pm^2 = \tfrac{1}{4}(I_4 \pm 2K + K^2) = \tfrac{1}{2}(I_4 \pm K) = P_\pm,$$

and $P_+P_- = \frac{1}{4}(I_4 - K^2) = 0$; the reversed product is the same. Their adjoints equal themselves and their sum is $I_4$. Also $K(I_4 \pm K) = K \pm I_4 = \pm(I_4 \pm K)$, proving $HP_\pm = \pm EP_\pm$. Each displayed Clifford matrix has trace zero. Hence tr $H = 0$ and tr $P_\pm = 2$. An idempotent splits the space into its range and kernel, acting as identity and zero respectively. Its trace is therefore its rank, proving rank two.

The identity $I_4 = P_+ + P_-$ and the two eigenvalue identities give, by induction, $H^k = E^k P_+ + (-E)^k P_-$ for every $k \geq 0$. Substitute into the absolutely convergent series to obtain the spectral expression. Expanding its projectors gives the cosine and sine expression.

When $E = 0$, the sum $m^2 + |p|^2$ of nonnegative squares is zero. Therefore $m = 0, p = 0$, and $H = 0$. Its exponential is $I_4$. The quotient $\sin(tE)/E$ tends to $t$ at zero; multiplying its filled value by $H = 0$ gives the same identity. No division by zero is used. □

The proof applies unchanged to $H_{h,n}$, with energy $E_{h,n}$. A positive-energy continuum mode can have $E_{h,n} = 0$ at a nonzero spacing when $m = 0$ and all centered components vanish. That mode uses the zero-energy branch. Positivity of the continuum energy does not license division by the centered energy at every spacing.

**Proposition PH04 (group law, unitarity, and stability).** For real $s, t$,

$$U_H(0) = I_4, \qquad U_H(s+t) = U_H(s)U_H(t),$$

$$U_H(-t) = U_H(t)^{-1} = U_H(t)^*, \qquad U_H(t)^*U_H(t) = U_H(t)U_H(t)^* = I_4.$$

Consequently,

$$\langle U_H(t)v, U_H(t)w\rangle = \langle v, w\rangle, \qquad \|U_H(t)v\|_2 = \|v\|_2$$

for all spinors. These statements hold for both mode Hamiltonians at every real $m, h, t$.

*Proof.* When $E > 0$, the spectral form and orthogonality of $P_\pm$ give

$$U_H(s)U_H(t) = e^{-i(s+t)E}P_+ + e^{i(s+t)E}P_- = U_H(s+t).$$

At zero time the sum is $I_4$. Hermiticity of the projectors and reality of $tE$ give $U_H(t)^* = e^{itE}P_+ + e^{-itE}P_- = U_H(-t)$. Setting $s = -t$ proves the inverse identities. When $E = 0$, every assertion follows from $U_H(t) = I_4$.

For any matrix $A$, expansion and interchange of two finite sums gives $\langle Av, w\rangle = \langle v, A^*w\rangle$. Thus

$$\langle U_H(t)v, U_H(t)w\rangle = \langle v, U_H(t)^*U_H(t)w\rangle = \langle v, w\rangle.$$

Take $w = v$ and then nonnegative square roots to obtain norm preservation. PH02 supplies the mode cases. □

An equivalent bounded-operator argument is represented in the formal source: $-itH$ is skew-adjoint, and its exponential is unitary. Our spectral proof supplies additional explicit projectors and formulas.

## 6 Fourier orthogonality and packet conservation

Let $J_n : \mathbb{C}^4 \to L^2(\mathbb{T}^3;\mathbb{C}^4)$ be $J_n v = e_n v$.

**Proposition PH05 (mode isometry and finite Parseval identity).** For modes $n, k$ and spinors $v, w$,

$$\langle J_n v, J_k w\rangle_{L^2} = \begin{cases} \langle v, w\rangle, & n = k, \\ 0, & n \neq k. \end{cases}$$

In particular, $J_n$ is a linear isometry and

$$\left\| \sum_{n\in S} J_n a_n \right\|_{L^2}^2 = \sum_{n\in S} \|a_n\|_2^2.$$

*Proof.* The constant spinors factor out of the scalar integral:

$$\langle J_n v, J_k w\rangle_{L^2} = \langle v, w\rangle \int_{\mathbb{T}^3} e^{2\pi i(k-n)\cdot x}\, dx.$$

The integral is the product of three one-dimensional integrals. For integer $r$, the integral of $e^{2\pi i r x}$ over $[0,1)$ is $1$ when $r = 0$; for $r \neq 0$, it is $(e^{2\pi i r} - 1)/(2\pi i r) = 0$. The product is therefore $1$ exactly when $n = k$. Linearity is immediate, and setting $k = n, w = v$ proves isometry. Expansion of the squared norm of the finite sum gives a double sum of pairings. The off-diagonal pairings vanish and the diagonal pairings are $\|a_n\|_2^2$. □

Define the evolved packets

$$f(t,x) = \sum_{n\in S} e_n(x) U_{H_n}(t) a_n, \qquad f_h(t,x) = \sum_{n\in S} e_n(x) U_{H_{h,n}}(t) a_n.$$

Both remain in $\mathcal{F}_S$. Their operators $U_S(t), U_{h,S}(t)$ have the finite block diagonal coefficient actions with blocks $U_{H_n}(t), U_{H_{h,n}}(t)$.

**Theorem PH06 (exact finite-packet norm conservation).** For every real $m, h, t$, finite $S$, and coefficient family $a$,

$$\|f(t)\|_{L^2}^2 = \|f_h(t)\|_{L^2}^2 = \sum_{n\in S} \|a_n\|_2^2 = \|f_a\|_{L^2}^2.$$

Both packet evolutions preserve the $L^2$ inner product between two packets in $\mathcal{F}_S$, and each forms a one-parameter unitary group there.

*Proof.* Apply PH05 to the evolved coefficients and PH04 on each mode:

$$\|f_h(t)\|_{L^2}^2 = \sum_{n\in S} \|U_{H_{h,n}}(t) a_n\|_2^2 = \sum_{n\in S} \|a_n\|_2^2.$$

The continuum argument is identical. For coefficient families $a, b$, the same two propositions give

$$\langle f_{h,a}(t), f_{h,b}(t)\rangle_{L^2} = \sum_{n\in S} \langle U_{H_{h,n}}(t) a_n, U_{H_{h,n}}(t) b_n\rangle = \sum_{n\in S} \langle a_n, b_n\rangle.$$

Composition of coefficient blocks proves the packet group law, and inverse time proves surjectivity. This establishes unitarity on $\mathcal{F}_S$ using only finite sums. □

The quantity conserved is squared Hilbert norm, or total probability for a normalized state. The parameter $m$, the positive shell value $E_n$, and Hamiltonian expectation are different quantities. Calling squared norm a packet mass does not identify it with the rest mass or physical Hamiltonian energy.

## 7 Generator consistency and dispersion error

The mode Hamiltonian difference contains no mass term:

$$\Delta_{h,n} := H_{h,n} - H_n = \sum_{j=x,y,z} \delta_j(h,n)\alpha_j, \qquad \delta_j(h,n) = q_j(h,p(n)) - p_j(n).$$

Define

$$C(n) = \frac{1}{6}\left(\sum_{j=x,y,z} |p_j(n)|^6\right)^{1/2}.$$

This constant depends on the fixed mode and increases cubically with momentum scale. That dependence is essential in a fixed-cutoff result.

**Proposition PH07 (quadratic generator consistency).** For every real $h$ and fixed mode $n$,

$$\|\Delta_{h,n}\| = |\delta(h,n)| \le h^2 C(n).$$

For every fixed finite packet,

$$\left\|\sum_{n\in S} e_n \Delta_{h,n} a_n\right\|_{L^2}^2 = \sum_{n\in S} |\delta(h,n)|^2 \|a_n\|_2^2,$$

and

$$\left\|\sum_{n\in S} e_n \Delta_{h,n} a_n\right\|_{L^2} \le h^2 \left(\sum_{n\in S} C(n)^2 \|a_n\|_2^2\right)^{1/2}.$$

*Proof.* First prove the scalar estimate without a sign restriction on $h$. The fundamental theorem of calculus gives $|\sin u| \le |u|$, since $\sin u = \int_0^u \cos v\, dv$ and $|\cos v| \le 1$. Therefore

$$|1 - \cos u| = 2\sin^2(u/2) \le u^2/2.$$

For $z \ge 0$, integrate this bound in $\sin z - z = \int_0^z (\cos u - 1)\, du$ to obtain $|\sin z - z| \le z^3/6$. Oddness of the residual proves $|\sin z - z| \le |z|^3/6$ for negative $z$. For $h \ne 0$, substitute $z = hp_j(n)$ and divide by $|h|$:

$$|\delta_j(h,n)| \le \frac{h^2 |p_j(n)|^3}{6}.$$

For $h = 0$, the residual is zero by definition and the bound still holds. Squaring and adding proves $|\delta(h, n)| \le h^2 C(n)$.

The Hermitian matrix $\Delta_{h,n}$ satisfies $\Delta_{h,n}^2 = |\delta(h, n)|^2 I_4$ by PH01. Hence

$$\|\Delta_{h,n} v\|_2^2 = \langle v, \Delta_{h,n}^* \Delta_{h,n} v\rangle = |\delta(h, n)|^2 \|v\|_2^2.$$

Its induced operator norm is exactly $|\delta(h, n)|$: the upper bound holds on every vector and equality holds on every unit vector. Apply PH05 to the residual coefficients for the packet equality, substitute the scalar bounds, and take nonnegative square roots. □

The checked generator results state corresponding $O(h^2)$ estimates as $h \to 0^+$. The explicit constant and all-real inequality above are traditional deductions. Precisely, $r(h) = O(h^2)$ means that some $M, \varepsilon > 0$ satisfy $\|r(h)\| \le Mh^2$ for $0 < h < \varepsilon$. Fixed support and amplitudes permit finite summation of such bounds; no estimate uniform over all integer modes follows.

The reverse triangle inequality in $\mathbb{R}^4$, applied to $(m, q_h(n))$ and $(m, p(n))$, gives

$$|E_{h,n} - E_n| \le |q_h(n) - p(n)| \le h^2 C(n).$$

Moreover, $|\sin(hp_j)/h| \le |p_j|$ for $h \neq 0$, and the filled value is equal at zero. Thus $0 \le E_{h,n} \le E_n$. For a fixed mode, Taylor's formula gives

$$q_j(h, p(n)) = p_j(n) - \frac{h^2 p_j(n)^3}{6} + O(h^4),$$

$$E_{h,n}^2 = E_n^2 - \frac{h^2}{3} \sum_j p_j(n)^4 + O(h^4).$$

For clarity, Taylor's theorem through degree four has zero even coefficients for sine, and its fifth-derivative remainder is at most $|hp_j|^5/120$. Division by $|h|$ gives $O(h^4)$; squaring the three expansions and summing preserves that order for a fixed mode. If $E_n > 0$, the square root is smooth near $E_n^2$, so

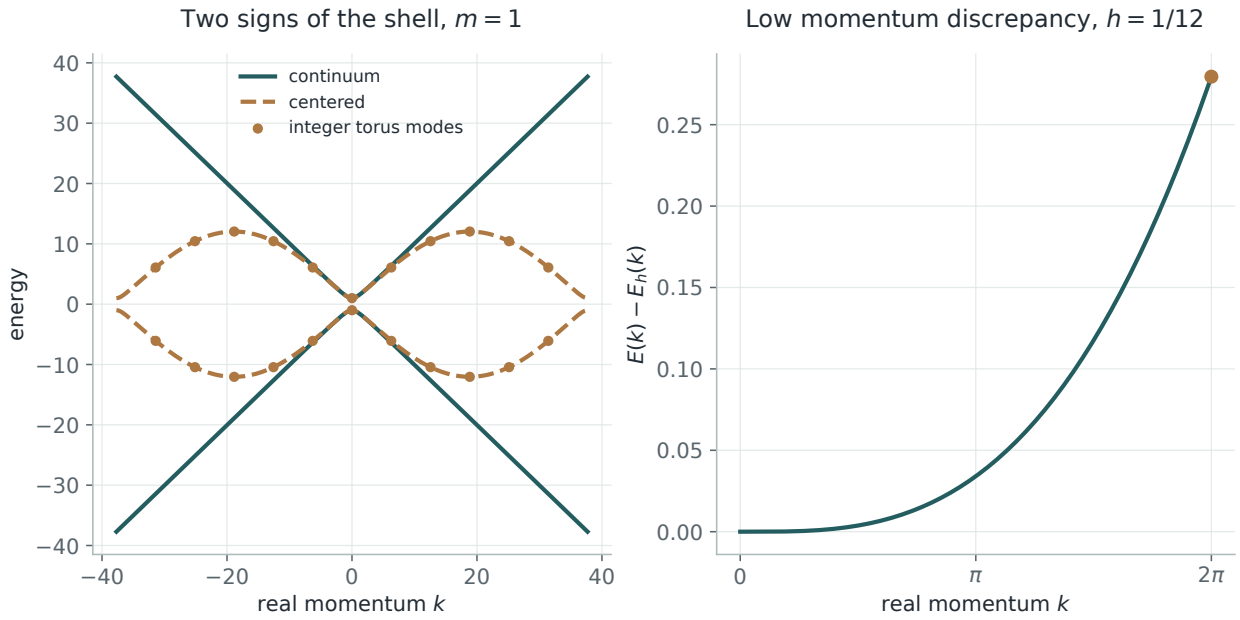
$$E_{h,n} = E_n - \frac{h^2}{6E_n} \sum_j p_j(n)^4 + O(h^4).$$

If $E_n = 0$, then $m = 0, n = 0$ and $E_{h,n} = 0$ exactly for every $h$. The expansion containing $1/E_n$ is not used.

For fixed time, the phase error on either energy branch obeys

$$|e^{\pm itE_{h,n}} - e^{\pm itE_n}| \le |t|\,|E_{h,n} - E_n|,$$

by integrating the derivative of $s \mapsto e^{\pm its}$ between the two real energies. Phase comparison alone does not compare full spinor evolution: the eigenspaces also depend on the kinetic vector. The operator argument below controls both effects.




The continuum and centered shell branches along one momentum axis, with $m = 1$, $h = 1/12$. Marked points are integer torus modes; intermediate points use the real-symbol extension.

## 8 Fixed-time propagator consistency

**Theorem PH08 (fixed-mode and fixed-packet convergence).** For fixed real $m, t$ and fixed $n$,

$$U_{H_{h,n}}(t) - U_{H_n}(t) = O(h^2) \quad (h \to 0^+).$$

For fixed finite $S$ and amplitudes $a$,

$$f_h(t) - f(t) = O(h^2) \quad \text{in } L^2(\mathbb{T}^3; \mathbb{C}^4),$$

and in particular $\|f_h(t) - f(t)\|_{L^2} \to 0$. The exact residual identity is

$$\|f_h(t) - f(t)\|_{L^2}^2 = \sum_{n \in S} \|(U_{H_{h,n}}(t) - U_{H_n}(t))a_n\|_2^2.$$

*Proof.* Give a fixed-time proof independent of the later compact-time extension. For arbitrary matrices $X, Y$, telescoping products give

$$X^k - Y^k = \sum_{r=0}^{k-1} X^{k-1-r}(X - Y)Y^r \quad (k \geq 1).$$

No commutation is assumed. Expanding the summands gives successive differences $X^{k-r}Y^r - X^{k-1-r}Y^{r+1}$, which cancel. If $\|X\|, \|Y\| \leq R$, then

$$\|X^k - Y^k\| \leq kR^{k-1}\|X - Y\|.$$

Subtract the absolutely convergent exponential series and sum:

$$\|e^X - e^Y\| \le \|X - Y\| \sum_{k=1}^{\infty} \frac{kR^{k-1}}{k!} = e^R \|X - Y\|.$$

PH02 gives $\|H_n\| = E_n, \|H_{h,n}\| = E_{h,n}$, by the norm-squared argument of PH07. Since $E_{h,n} \le E_n$, take $X = -itH_{h,n}$, $Y = -itH_n$, and $R = |t|E_n$. The result is

$$\|U_{H_{h,n}}(t) - U_{H_n}(t)\| \le |t| e^{|t|E_n} h^2 C(n).$$

This proves the modewise order, including $t = 0$. The difference of packets is the finite sum of the mode residuals, so PH05 gives its exact residual identity. Thus

$$\|f_h(t) - f(t)\|_{L^2} \le h^2 \left( \sum_{n \in S} |t|^2 e^{2|t|E_n} C(n)^2 \|a_n\|_2^2 \right)^{1/2}.$$

The constant is finite for fixed $t, S, a$, proving the order and convergence. □

The formal algebraic-coordinate proof obtains this order from continuity and local differentiability of the Banach-algebra exponential. Its operator norm belongs to the algebraic coordinate spinor type. The coefficient conversion in its packet definitions is continuous, so its checked torus-$L^2$ convergence is genuine. Our Euclidean PH08 also has the direct proof above. The following bridge explains why its coordinates agree with the separately defined Hilbert propagators.

## 9 The algebraic-to-Hilbert coordinate bridge

Let $\mathcal{A}$ be the coordinate space $\{0, 1, 2, 3\} \to \mathbb{C}$ with maximum norm $\|v\|_\infty = \max_r |v_r|$, and let $\mathcal{H}$ be the same coordinates with the Euclidean inner product. The map

$$T : \mathcal{A} \longrightarrow \mathcal{H}, \qquad (Tv)_r = v_r$$

is complex-linear and bijective; its inverse also retains the coordinates. It is not generally an isometry between these norms. The inequalities

$$\|v\|_\infty \le \|Tv\|_2 \le 2\|v\|_\infty$$

show that both directions are continuous. The first follows because each coordinate square is bounded by the sum; the second because each of four squares is at most $\|v\|_\infty^2$.

**Proposition PH09 (coordinate conjugacy of generators and exponentials).** Let $A$ be a complex-linear endomorphism of $\mathcal{A}$, and let $A^{\mathcal{H}} = TAT^{-1}$. For every complex $z$,

$$e^{zA^{\mathcal{H}}} = Te^{zA}T^{-1}.$$

For the source Clifford actions, the continuum and centered Hilbert generators are exactly these conjugates of their algebraic generators. Their exact propagators therefore have the same coordinates after applying $T$. The algebraic-coordinate packet, transported coefficientwise by $T$, agrees with the Hilbert packet for the corresponding amplitudes.

*Proof.* Induction gives $(TAT^{-1})^k = TA^kT^{-1}$. For $k = 0$, use $TT^{-1} = I$; for the induction step, cancel the adjacent $T^{-1}T$. Every exponential partial sum therefore satisfies

$$\sum_{k=0}^{N} \frac{z^k (A^{\mathcal{H}})^k}{k!} = T \left( \sum_{k=0}^{N} \frac{z^k A^k}{k!} \right) T^{-1}.$$

The exponential series converges in both operator norms. Conjugation is continuous: writing induced norms in the relevant spaces gives

$$\|TBT^{-1}\|_2 \le \|T\| \, \|B\|_\infty \, \|T^{-1}\| \le 2\|B\|_\infty.$$

Pass to the limit to obtain the identity.

The source's transported Clifford action is defined by $v \mapsto T(A(T^{-1}v))$. The same is true for the displayed diagonal beta action. Conjugation preserves real coefficients and finite sums, proving the Hamiltonian equality. Apply the exponential identity with $z = -it$. Substituting each coefficient equality into the finite Fourier sum then proves equality of the packets. □

The norm bound also transports quadratic operator errors between realizations. The bridge is a traditional proof in this paper. The checked source contains inverse coordinate identities and the pairing identification, but no complete generator-and-exponential conjugacy theorem. PH09's conjugacy itself remains an ordinary proof in this text. Blocks 44 and 45 separately establish Hilbert stability and consistency directly.

## 10 An exact three-mode packet

**Example PH10 (axis and diagonal modes at $h = 1/12$).** Set $m = 1$, $h = 1/12$, and

$$S = \{n_0, n_A, n_D\}, \qquad n_0 = (0,0,0), \quad n_A = (1,0,0), \quad n_D = (1,1,1).$$

Let $e_0 = (1,0,0,0)^T$ denote the first spinor basis vector, and choose

$$a_{n_0} = \frac{1}{\sqrt{2}} e_0, \qquad a_{n_A} = \frac{1}{2} e_0, \qquad a_{n_D} = \frac{1}{2} e_0.$$

Their squared norms are $1/2, 1/4, 1/4$, so the packet is normalized. Its continuum and centered momenta are

| Mode | $p(n)$ | $q_h(n)$ |
|---|---|---|
| $n_0$ | $(0,0,0)$ | $(0,0,0)$ |
| $n_A$ | $(2\pi,0,0)$ | $(6,0,0)$ |
| $n_D$ | $(2\pi,2\pi,2\pi)$ | $(6,6,6)$ |

Each nonzero component has $hp_j = \pi/6$, so $\sin(hp_j)/h = (1/2)/(1/12) = 6$. Thus

| Mode | $E_n$ | $E_{h,n}$ |
|---|---|---|
| $n_0$ | $1$ | $1$ |
| $n_A$ | $\sqrt{1+4\pi^2}$ | $\sqrt{37}$ |
| $n_D$ | $\sqrt{1+12\pi^2}$ | $\sqrt{109}$ |

These are exact symbolic equalities. The centered component $6$ is distinct from the integer mode label $1$ and continuum momentum $2\pi$.

For any real kinetic vector $r$, the exact matrix gives

$$H(1,r)e_0 = (1,0,r_z,r_x+ir_y)^T.$$

Set $E_r = \sqrt{1+|r|^2}$, $c_r(t) = \cos(E_r t)$, and $s_r(t) = \sin(E_r t)/E_r$. PH03 gives

$$u_r(t) = U_{H(1,r)}(t)e_0 = \begin{pmatrix} c_r(t) - is_r(t) \\ 0 \\ -ir_z s_r(t) \\ (r_y - ir_x)s_r(t) \end{pmatrix}.$$

In particular, $u_0(t) = e^{-it}e_0$. For the centered axis and diagonal modes,

$$u_A(t) = \begin{pmatrix} \cos(\sqrt{37}t) - i\sin(\sqrt{37}t)/\sqrt{37} \\ 0 \\ 0 \\ -6i\sin(\sqrt{37}t)/\sqrt{37} \end{pmatrix},$$

$$u_D(t) = \begin{pmatrix} \cos(\sqrt{109}t) - i\sin(\sqrt{109}t)/\sqrt{109} \\ 0 \\ -6i\sin(\sqrt{109}t)/\sqrt{109} \\ 6(1-i)\sin(\sqrt{109}t)/\sqrt{109} \end{pmatrix}.$$

The complete centered packet is

$$f_h(t,x) = \frac{e^{-it}}{\sqrt{2}}e_0 + \frac{1}{2}e^{2\pi i x_1}u_A(t) + \frac{1}{2}e^{2\pi i(x_1+x_2+x_3)}u_D(t).$$

The continuum packet has the same form with $u_A, u_D$ replaced by $u_{(2\pi,0,0)}$ and $u_{(2\pi,2\pi,2\pi)}$. At $t=0$, both equal

$$f_a(x) = \left(\frac{1}{\sqrt{2}} + \frac{1}{2}e^{2\pi i x_1} + \frac{1}{2}e^{2\pi i(x_1+x_2+x_3)}\right)e_0.$$

The spinor formula verifies stability directly:

$$\|u_r(t)\|_2^2 = c_r(t)^2 + s_r(t)^2(1+r_z^2+r_x^2+r_y^2) = \cos^2(E_r t) + \sin^2(E_r t) = 1.$$

Here $|c-is|^2 = c^2+s^2$ and $|r_y - ir_x|^2 = r_x^2 + r_y^2$, since the coefficients are real. Fourier orthogonality gives

$$\|f_h(t)\|_{L^2}^2 = \tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{4} = 1, \qquad \|f(t)\|_{L^2}^2 = 1.$$

This is an integrated statement. The initial pointwise density is not constant: at $x = 0$, it is $(1 + 1/\sqrt{2})^2 = 3/2 + \sqrt{2}$. Cross terms can change local density while their torus integrals vanish. Normalization does not mean that each point has density one.

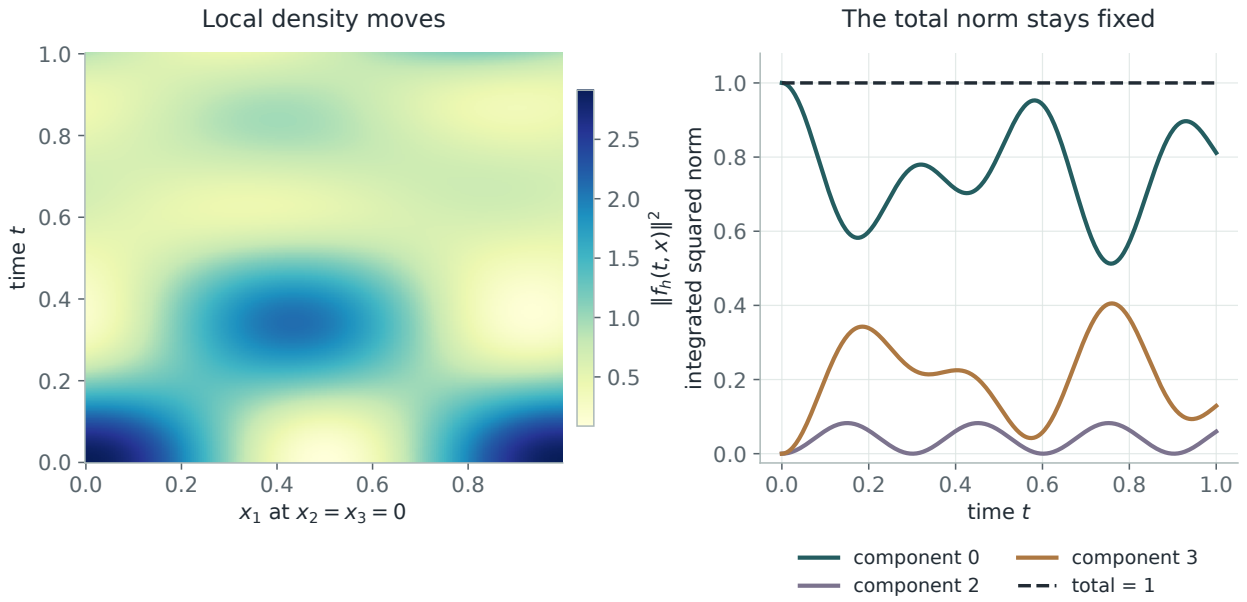
The error also has an exact expression. For fixed mode, let $r = p(n)$, $q = q_h(n)$, with the corresponding $c_r, s_r, c_q, s_q$. Coordinate subtraction gives

$$\|u_q(t) - u_r(t)\|_2^2 = (c_q - c_r)^2 + (s_q - s_r)^2 + \sum_{j=x,y,z} (q_j s_q - r_j s_r)^2.$$

For the selected packet, the zero mode contributes no error, so

$$\begin{aligned}\|f_h(t) - f(t)\|_{L^2}^2 = \frac{1}{4}\|u_{(6,0,0)}(t) - u_{(2\pi,0,0)}(t)\|_2^2 \\ + \frac{1}{4}\|u_{(6,6,6)}(t) - u_{(2\pi,2\pi,2\pi)}(t)\|_2^2.\end{aligned}$$

A plot or numerical evaluation illustrates this exact expression; conservation and normalization have already been proved symbolically.




The centered three-mode packet at $m = 1$, $h = 1/12$ changes local density and spinor composition while keeping total squared norm one. The image evaluates the closed formula at finitely many times.

## 11 A finite-matrix compact-time extension

The following proof gives an explicit Duhamel bound, including its sharp coefficient. This is an ordinary mathematical deduction with a complete proof. The formal compact-time convergence argument uses exponential continuity and is identified separately in the verification appendix.

**Lemma (finite Hermitian Duhamel estimate).** Let $A, B$ be Hermitian matrices on a nonzero finite-dimensional complex Hilbert space. For every real $t$,

$$\|e^{-itA} - e^{-itB}\| \le |t|\,\|A - B\|.$$

The same inequality holds trivially on the zero space.

*Proof.* First justify differentiation. On $|s| \le R$, the series for $e^{-isB}$ converges uniformly, and its derivative series is norm-dominated by

$$\sum_{k=1}^{\infty} \frac{kR^{k-1}\|B\|^k}{k!} = \|B\| e^{R\|B\|}.$$

Termwise differentiation therefore gives

$$\frac{d}{ds} e^{-isB} = -iBe^{-isB} = -ie^{-isB}B.$$

The same holds for $A$. Since $A$ commutes with every power of itself, it commutes with $e^{-iuA}$. Fix $t$ and define $F(s) = e^{-i(t-s)A}e^{-isB}$. The product rule gives

$$F'(s) = ie^{-i(t-s)A}(A - B)e^{-isB}.$$

The endpoints are $F(0) = e^{-itA}$ and $F(t) = e^{-itB}$. Integrating entrywise in the finite matrix space gives the oriented-integral identity

$$e^{-itB} - e^{-itA} = i \int_0^t e^{-i(t-s)A}(A - B)e^{-isB}\, ds.$$

For Hermitian matrices the two exponentials in the integrand are unitary. This does not rely on the special Clifford algebra: the series gives $(e^{-iuA})^* = e^{iuA}$, and the Cauchy product of the two absolutely convergent series gives $e^{iuA}e^{-iuA} = I$. Indeed, the coefficient of $A^k$ for $k > 0$ is $(iu)^k \sum_{r=0}^{k} \binom{k}{r}(-1)^r/k! = 0$, and the constant coefficient is $I$. A unitary matrix on a nonzero space has norm one. Submultiplicativity therefore bounds the integrand norm by $\|A - B\|$. For $t \ge 0$, integration gives $t\|A - B\|$; for $t < 0$, reversing the interval gives $(-t)\|A - B\|$. This proves the estimate for every real time. □

**Theorem PH11 (time-explicit finite-packet bound).** For every real $m, h, t$, fixed mode $n$, and fixed finite packet,

$$\|U_{H_{h,n}}(t) - U_{H_n}(t)\| \le |t|\,|\delta(h,n)| \le |t|h^2 C(n),$$

$$\begin{aligned}\|f_h(t) - f(t)\|_{L^2} &\le |t| \left(\sum_{n\in S} |\delta(h,n)|^2 \|a_n\|_2^2\right)^{1/2} \\ &\le |t|h^2 \left(\sum_{n\in S} C(n)^2 \|a_n\|_2^2\right)^{1/2}.\end{aligned}$$

Consequently, for every $T \ge 0$,

$$\sup_{|t|\le T} \|f_h(t) - f(t)\|_{L^2} \le Th^2 \left(\sum_{n\in S} C(n)^2 \|a_n\|_2^2\right)^{1/2},$$

and evolution converges uniformly on every compact time interval for fixed support and initial amplitudes.

*Proof.* Apply the Duhamel lemma with $A = H_{h,n}$, $B = H_n$. Both are Hermitian by PH02, and PH07 gives the residual norm and bound. For each amplitude,

$$\|(U_{H_{h,n}}(t) - U_{H_n}(t))a_n\|_2 \le |t|\,|\delta(h,n)|\,\|a_n\|_2 \le |t|h^2 C(n)\|a_n\|_2.$$

Square and sum over $S$, apply the exact residual Parseval identity of PH08, and take the nonnegative square root. The middle coefficient is exactly the initial generator-residual norm in PH07 because these Clifford differences satisfy $\Delta_{h,n}^* \Delta_{h,n} = |\delta(h,n)|^2 I_4$. This identification is specific to the displayed symbol differences; the general Duhamel lemma is an operator-norm estimate. Replacing $|t|$ by $T$ gives the compact-time bound. Its right side tends to zero as $h \to 0$. □

For PH10, $C(n_0) = 0$, $C(n_A) = (2\pi)^3/6$, and $C(n_D) = \sqrt{3}(2\pi)^3/6$, so

$$\left(\sum_{n\in S} C(n)^2 \|a_n\|_2^2\right)^{1/2} = \frac{(2\pi)^3}{6} = \frac{4\pi^3}{3}.$$

At its spacing, put $d = 2\pi - 6 > 0$. The axis residual has magnitude $d$, the diagonal residual has magnitude $\sqrt{3}d$, and their squared amplitude weights are $1/4, 1/4$. Hence the exact middle coefficient is

$$\left(\tfrac{1}{4}d^2 + \tfrac{1}{4}\,3d^2\right)^{1/2} = d,$$

and the worked bound is

$$\|f_{1/12}(t) - f(t)\|_{L^2} \le (2\pi - 6)|t| \le \frac{\pi^3}{108}|t|.$$

The triangle inequality also gives $\|f_h(t) - f(t)\|_{L^2} \le 2$ for this normalized packet, so one may use the smaller bound. Neither says that a fixed nonzero spacing has small error for arbitrarily large time. Nor does the time bound remove the cutoff: its coefficient contains high powers of mode momenta. For

arbitrary $L^2$ data, the Fourier-tail argument in the following part provides convergence without this uniform quadratic coefficient.

## 12 Probability, Hamiltonian expectation, and physical scope

Let $H_S, H_{h,S}$ denote the finite block diagonal Hamiltonians. In coefficient coordinates, squared norm and continuum Hamiltonian expectation are

$$N(a) = \sum_{n\in S} \|a_n\|_2^2, \qquad \mathcal{E}(a) = \sum_{n\in S} \langle a_n, H_n a_n\rangle,$$

with centered expectation $\mathcal{E}_h(a)$ obtained by replacing $H_n$ with $H_{h,n}$. Hermiticity makes each expectation real: its complex conjugate is $\langle H_n a_n, a_n\rangle = \langle a_n, H_n a_n\rangle$.

**Proposition PH12 (separate conservation of Hamiltonian expectation).** Each exact packet evolution preserves the expectation of its own time-independent Hamiltonian:

$$\mathcal{E}(U_S(t)a) = \mathcal{E}(a), \qquad \mathcal{E}_h(U_{h,S}(t)a) = \mathcal{E}_h(a).$$

Here the packet operators act on coefficients through the isometry $\mathcal{K}_S \cong \mathcal{F}_S$. Squared-norm conservation does not identify either expectation with $N(a)$. At common initial data,

$$|\mathcal{E}_h(a) - \mathcal{E}(a)| \le h^2 \sum_{n\in S} C(n)\|a_n\|_2^2.$$

*Proof.* The exponential of a fixed matrix commutes with that matrix, term by term in its power series. Unitarity gives on each mode

$$\langle U_H(t)a, HU_H(t)a\rangle = \langle a, U_H(t)^* H U_H(t)a\rangle = \langle a, Ha\rangle.$$

Sum to prove both conservation laws. For common initial coefficients, Cauchy–Schwarz and PH07 give

$$|\langle a_n, (H_{h,n} - H_n)a_n\rangle| \le \|a_n\|_2 \|\Delta_{h,n} a_n\|_2 \le h^2 C(n)\|a_n\|_2^2.$$

The finite-sum triangle inequality proves the comparison. □

A unit vector in the negative-energy range of $P_-$ has squared norm one but expectation $-E$; a unit vector in the positive-energy range has expectation $E$. Norm and energy are therefore different even for one mode. For $e_0$, the matrix gives $\langle e_0, H(m,p)e_0\rangle = m$, since the kinetic terms have zero diagonal entry in that component. PH10 has expectation $1$ for both Hamiltonians. Its values $\sqrt{37}, \sqrt{109}$ are shell magnitudes, and its spinors contain both energy signs. Their expectations need not equal a positive shell value.

The analysis is free and linear. No gauge field, forcing, interaction, or time-dependent Hamiltonian occurs in these symbols. Other project modules treat different transport structures; importing them does not convert these statements into theorems about interacting fields. The centered propagator is the exact exponential of a modified spatial symbol. A numerical approximation to that exponential must be checked separately for its own norm error.

# Completion and continuum evolution

## 13 The complete coefficient Hilbert space

Let

$$\mathcal{K} = \ell^2(\mathbb{Z}^3; \mathbb{C}^4) = \left\{ a = (a_n) : \sum_{n \in \mathbb{Z}^3} \|a_n\|^2 < \infty \right\}.$$

Its inner product is $\langle a, b \rangle = \sum_n \langle a_n, b_n \rangle$. Absolute convergence follows from the scalar Cauchy–Schwarz inequality. The norm therefore measures the total squared amplitude across all Fourier packets. A finite family of amplitudes embeds by assigning zero to every absent mode.

For a finite set $S \subset \mathbb{Z}^3$, define the truncation $P_S a$ by retaining $a_n$ when $n \in S$ and assigning zero otherwise. Then

$$\|a - P_S a\|^2 = \sum_{n \notin S} \|a_n\|^2 .$$

The sum on the right can be made arbitrarily small because the full nonnegative series converges. Thus finite packets are dense in $\mathcal{K}$. The statement concerns the norm of the omitted amplitudes and does not depend on a particular ordering of the integer lattice.

**Theorem 13.1** (Completion of a modewise isometry)**.** *Suppose that for every mode $n$, $V_n : \mathbb{C}^4 \to \mathbb{C}^4$ is a complex linear isometry. The coordinate rule $(Va)_n = V_n a_n$ defines a complex linear isometry on $\mathcal{K}$. If each $V_n$ has an inverse isometry $W_n$, the induced $V$ and $W$ are inverse maps on $\mathcal{K}$.*

*Proof.* For each $n$, $\|V_n a_n\|^2 = \|a_n\|^2$. Summability of the latter terms gives membership of $Va$ in $\mathcal{K}$, and summing the equalities gives $\|Va\| = \|a\|$. Linearity holds coordinatewise. The two inverse identities hold on each coordinate and therefore on the coefficient fields themselves. □

Apply this construction to

$$V_n(t) = e^{-itH_n}, \qquad V_{h,n}(t) = e^{-itH_{h,n}}.$$

The finite-dimensional norm preservation already proved supplies its hypothesis. Define

$$(U(t)a)_n = V_n(t)a_n, \qquad (U_h(t)a)_n = V_{h,n}(t)a_n.$$

Both maps are everywhere-defined unitary maps on $\mathcal{K}$. Their inverses are obtained at time $-t$. The modewise time-group laws imply

$$U(s+t) = U(s)U(t), \qquad U_h(s+t) = U_h(s)U_h(t).$$

This construction uses the bounded unitary map on each mode. It does not apply an unbounded Hamiltonian to an arbitrary coefficient field. In particular, the evolution is defined for more initial states than the Hamiltonian action constructed below.

### 13.1 The invariant finite packets

Both families commute with every finite truncation:

$$U(t)P_S = P_S U(t), \qquad U_h(t)P_S = P_S U_h(t).$$

The equality follows directly from the diagonal mode action. It says that the full evolution restricts to the finite-packet evolution already constructed. It also says that the missing tail evolves without feeding new amplitude into the retained mode set in this free model.

This property would have to be reconsidered in a model whose coefficients couple different Fourier modes. The completion argument for a diagonal multiplier makes essential use of the simple resolved action, even though the general packet language also describes nondiagonal transfers.

## 14 Removing the Fourier cutoff

A finite-packet consistency theorem alone says nothing about an arbitrary field with infinitely many nonzero coefficients. The missing estimate must control the error introduced by truncating that field. Exact unitarity gives a bound whose tail terms do not grow with time.

**Lemma 14.1** (Two tails and a finite core)**.** *Let $V$ and $W$ be linear isometries on a normed space. For any $a$ and any approximation $b$,*

$$\|Va - Wa\| \le 2\,\|a - b\| + \|Vb - Wb\|\,.$$

*Proof.* Write the residual as $V(a-b)+(Vb-Wb)+W(b-a)$. The triangle inequality bounds its norm by the sum of the three norms. Each outer term has norm $\|a - b\|$ because its operator is an isometry. □

Take $V = U_h(t)$, $W = U(t)$, and $b = P_S a$. For a fixed finite $S$, the compact-time packet estimate gives

$$\|U_h(t)P_S a - U(t)P_S a\| \le K_{S,a,T} h^2$$

for all $|t| \le T$ and sufficiently small positive $h$. The constant is allowed to depend on the support, the amplitudes, and the time window. Combining the estimates gives

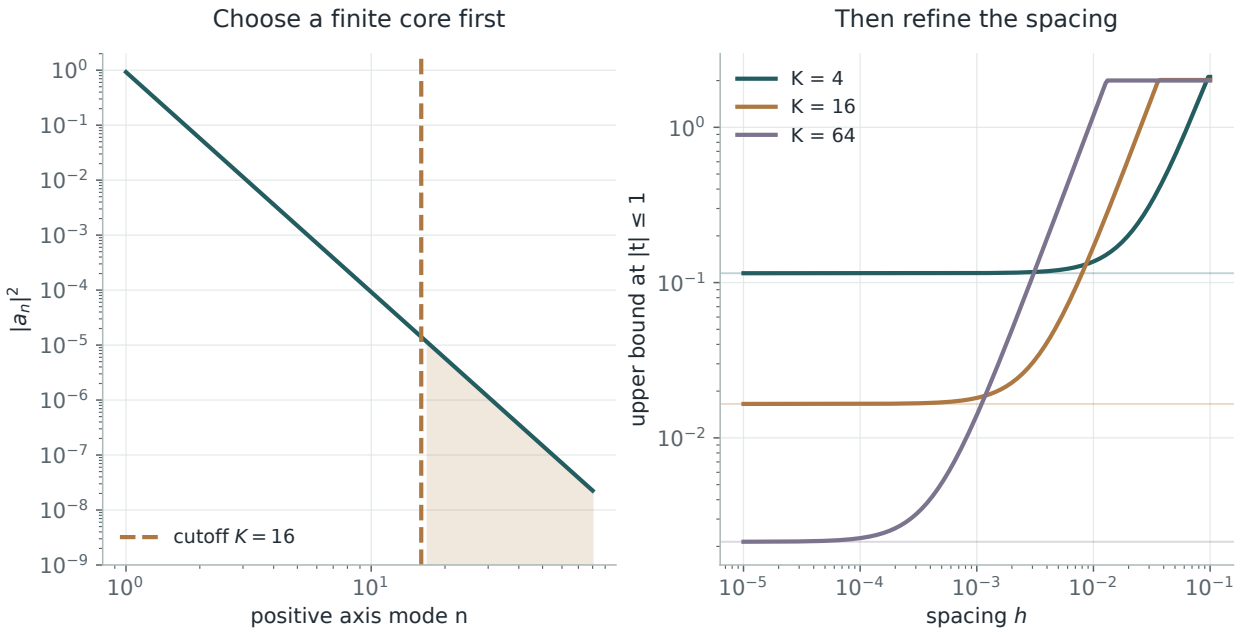
$$\|U_h(t)a - U(t)a\| \le 2\,\|a - P_S a\| + K_{S,a,T} h^2.$$

**Theorem 14.2** (Compact-time strong convergence)**.** *For every $a \in \mathcal{K}$, every $T \ge 0$, and every $\varepsilon > 0$, there exists $\delta > 0$ such that*

$$0 < h < \delta, \quad |t| \le T \quad \Longrightarrow \quad \|U_h(t)a - U(t)a\| < \varepsilon.$$

*Proof.* Choose a finite $S$ such that $\|a - P_S a\| < \varepsilon/4$. The finite-packet theorem gives a sufficiently small positive $\delta$ for which the core residual is less than $\varepsilon/2$ throughout the time interval. The two-tail inequality then makes the full residual less than $\varepsilon$. The choice of $S$ precedes the choice of $h$ and works for every time in the same interval. □

The quantifiers are important. One initial field and one compact time interval are fixed before the mesh scale is chosen. The theorem does not assert convergence in operator norm on the entire unit ball. It also does not assign a universal $O(h^2)$ rate to every $a \in \mathcal{K}$. For arbitrary square-summable data, a small Fourier tail may require a large cutoff, and the finite-core constant may grow with that cutoff.




The proof separates the approximation of the initial field from the consistency estimate on finitely many modes. The diagram records the order of choices in the argument.

## 15 Fourier synthesis on the physical torus

The scalar Fourier characters $e_n(x) = e^{2\pi i n\cdot x}$ are normalized relative to Haar measure of total mass one. For a four-spinor field, apply the scalar Fourier coefficient map to each of its four coordinates. The resulting map is

$$\mathcal{F} : \mathcal{H} \longrightarrow \mathcal{K}, \qquad \mathcal{H} = L^2(\mathbb{T}^3; \mathbb{C}^4).$$

The four-spinor Parseval identity and surjectivity of synthesis produce a complex linear isometric equivalence. In ordinary notation,

$$\|\mathcal{F} f\|_{\mathcal{K}} = \|f\|_{\mathcal{H}}, \qquad \mathcal{F}^{-1}\mathcal{F} f = f, \qquad \mathcal{F}\mathcal{F}^{-1} a = a.$$

The inverse is understood as convergence in $L^2$. A general field need not have a pointwise-convergent Fourier series, and no such convergence is required here.

For a finite-support coefficient field, the inverse is the familiar finite sum

$$\mathcal{F}^{-1} a = \sum_n e_n a_n.$$

The complete-space equivalence extends that reconstruction while retaining its exact norm. Its surjectivity is the step that rules out having built only a proper closed subspace of the physical field space.

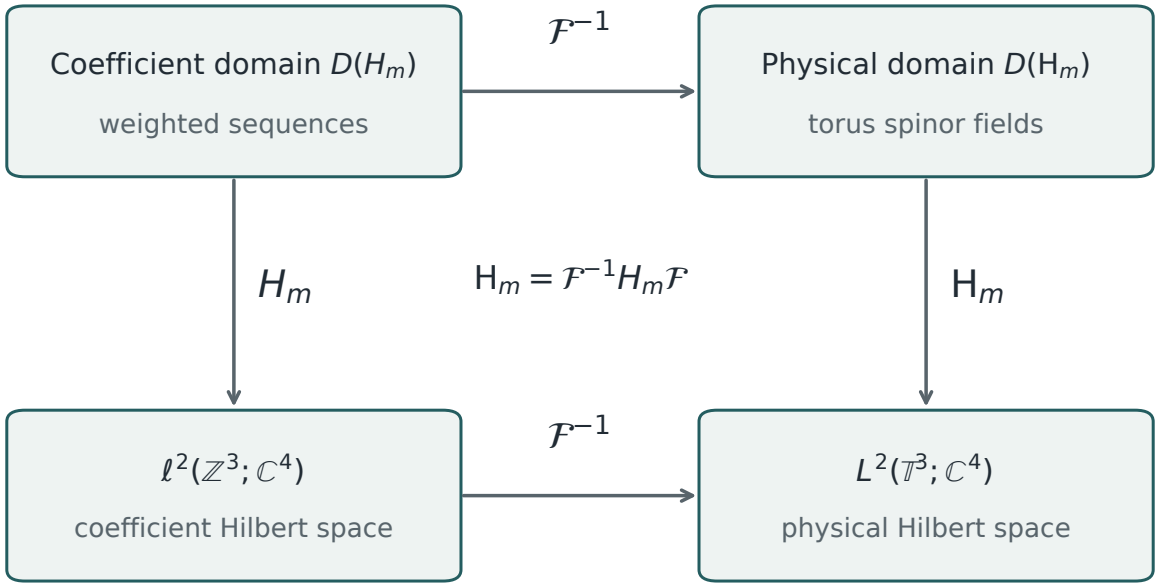
**Definition 15.1** (Physical propagators)**.** Define

$$\mathcal{U}(t) = \mathcal{F}^{-1} U(t) \mathcal{F}, \qquad \mathcal{U}_h(t) = \mathcal{F}^{-1} U_h(t) \mathcal{F}.$$

Both preserve the physical $L^2$ norm and inner product. Their group laws follow by inserting $\mathcal{F}\mathcal{F}^{-1} = I$ between consecutive factors. Their residual satisfies the exact equality

$$\|\mathcal{U}_h(t)f - \mathcal{U}(t)f\|_{\mathcal{H}} = \|U_h(t)\mathcal{F}f - U(t)\mathcal{F}f\|_{\mathcal{K}}\,.$$

The compact-time convergence theorem therefore transfers to every $f \in \mathcal{H}$ without an additional constant.




The horizontal maps preserve inner products and norms.

Fourier analysis and synthesis change the representation while preserving the Hilbert norm. The operator in each representation is related by conjugation.

The physical torus in this theorem is a fixed flat compact space. The word physical identifies the function-space realization of the fields, rather than establishing empirical adequacy or deriving a spacetime geometry. Periodicity is an explicit boundary condition. The next part constructs the Hamiltonian on exactly this space and retains that same boundary condition.

# The unbounded Hamiltonian

## 16 A domain defined by the operator output

The continuum mode energy grows with $|n|$. A square-summable sequence $a_n$ need not remain square summable after multiplication by $H_n$. This is why the full Hamiltonian is represented as a partial linear map with a specified domain.

Define

$$\mathcal{D}(\widehat{H}_m) = \left\{ a \in \mathcal{K} : \sum_n \|H_n a_n\|^2 < \infty \right\}, \qquad (\widehat{H}_m a)_n = H_n a_n.$$

The domain is maximal for this coefficient action: it contains precisely those coefficient fields for which the stated output belongs to the ambient Hilbert space. It is a vector subspace. For example, $\|x+y\|^2 \le 2\|x\|^2 + 2\|y\|^2$ and linearity of each $H_n$ prove closure under addition. Scalar multiplication preserves summability as well.

Every finite-support field lies in this domain, since the output again has finite support. Density of finite-support fields therefore proves density of $\mathcal{D}(\widehat{H}_m)$ in $\mathcal{K}$. Density matters because it gives uniqueness of an adjoint value when such a value exists.

The source represents this object with Mathlib's `LinearPMap`. The domain is a submodule and is carried by the operator itself. It is not replaced with a total function that assigns an arbitrary value to every vector outside the domain.

## 17 The adjoint domain and self-adjointness

Modewise Hermiticity gives symmetry on the maximal domain. If $a, b$ belong to that domain, the two series in

$$\left\langle \widehat{H}_m a, b \right\rangle = \sum_n \langle H_n a_n, b_n \rangle = \sum_n \langle a_n, H_n b_n \rangle = \left\langle a, \widehat{H}_m b \right\rangle$$

converge absolutely by Cauchy–Schwarz. This establishes the inclusion $\widehat{H}_m \subseteq \widehat{H}_m^*$, including the corresponding domain inclusion.

Equality requires the converse. A merely symmetric operator can have a strictly larger adjoint domain. Here the coordinate structure makes it possible to determine that domain directly.

**Theorem 17.1** (Self-adjoint maximal multiplier)**.** *The densely defined operator $\widehat{H}_m$ equals its Hilbert adjoint.*

*Proof.* Let $b \in \mathcal{D}(\widehat{H}_m^*)$ and write $c = \widehat{H}_m^* b$. By the adjoint relation,

$$\left\langle \widehat{H}_m a, b \right\rangle = \langle a, c \rangle \qquad (a \in \mathcal{D}(\widehat{H}_m)).$$

Fix a mode $n$ and choose $a$ with the single nonzero coefficient $a_n = v$, where $v \in \mathbb{C}^4$ is arbitrary. Such a field belongs to the maximal domain. The adjoint identity becomes

$$\langle H_n v, b_n \rangle = \langle v, c_n \rangle .$$

Hermiticity gives $\langle v, H_n b_n \rangle = \langle v, c_n \rangle$ for every $v$. Taking $v = H_n b_n - c_n$ shows $c_n = H_n b_n$. Because $c \in \mathcal{K}$, the sequence $(H_n b_n)_n$ is square summable. Thus $b \in \mathcal{D}(\widehat{H}_m)$ and the two operator actions agree there. This proves $\widehat{H}_m^* \subseteq \widehat{H}_m$. Symmetry supplies the reverse inclusion. □

The argument identifies the output before inferring membership in the maximal domain. It therefore rules out an extra adjoint-domain vector through a concrete summability requirement, rather than assuming the desired domain equality.

Closedness follows from self-adjointness. It can also be seen directly for this multiplier. If $a^{(r)} \to a$ and $\widehat{H}_m a^{(r)} \to b$ in $\mathcal{K}$, then convergence holds at each coordinate. Since the finite matrix $H_n$ is continuous,

$$b_n = \lim_r H_n a_n^{(r)} = H_n a_n.$$

The output sequence equals $b \in \mathcal{K}$, so $a$ belongs to the maximal domain and $\widehat{H}_m a = b$. This direct argument is included as an ordinary explanatory deduction.

## 18 Finite packets approximate the graph

For a domain vector $a$, truncation commutes with the diagonal multiplier:

$$\widehat{H}_m P_S a = P_S \widehat{H}_m a.$$

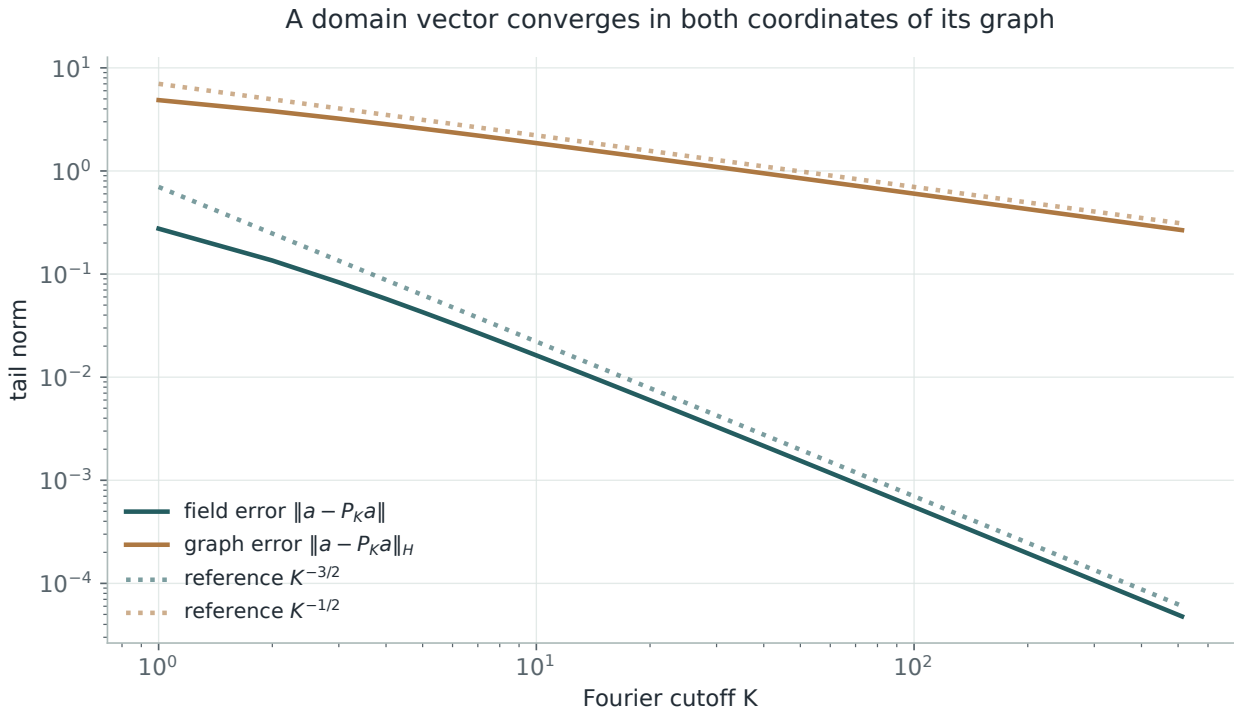
Both $a$ and $\widehat{H}_m a$ are square summable. Given $\varepsilon > 0$, choose finite sets that make each of their tails small and take their union. The resulting one finite set $S$ satisfies

$$\|a - P_S a\| < \varepsilon, \qquad \left\|\widehat{H}_m a - \widehat{H}_m P_S a\right\| < \varepsilon.$$

This graph-approximation statement is represented in the formal source. The finite Fourier fields are dense for the graph norm

$$\|a\|_{\text{graph}}^2 = \|a\|^2 + \left\|\widehat{H}_m a\right\|^2.$$

The term *graph core* describes this mathematical conclusion. The source proves the simultaneous approximation theorem without introducing a separate `IsCore` wrapper.




Approximating a field in its Hilbert norm and approximating it together with its Hamiltonian image are different requirements. The plotted family is a bounded illustration of the graph-tail calculation.

This result concerns truncations of one continuum operator. A lattice-to-continuum graph-convergence theorem compares different operators and still requires its own compatible-domain construction. The word graph in both phrases does not identify those two tasks.

## 19 The exact first-order Fourier domain

The Clifford square identity sharpens the definition of the maximal domain. Since $H_n^* = H_n$ and $H_n^2 = (m^2 + |p(n)|^2)I_4$,

$$\|H_n v\|^2 = \langle v, H_n^* H_n v\rangle = (m^2 + |p(n)|^2)\,\|v\|^2 .$$

For the coefficient field this gives the exact equivalence

$$a \in \mathcal{D}(\widehat{H}_m) \quad \Longleftrightarrow \quad \sum_n (m^2 + |p(n)|^2)\,\|a_n\|^2 < \infty.$$

The mass term is already summable because $a \in \mathcal{K}$. Membership is therefore equivalent to summability of $\sum_n |p(n)|^2\,\|a_n\|^2$.

**Theorem 19.1** (Weighted domain and mass independence). *For every real $m$,*

$$\mathcal{D}(\widehat{H}_m) = \left\{ a \in \mathcal{K} : \sum_n (1 + |p(n)|^2)\,\|a_n\|^2 < \infty \right\}.$$

*Consequently $\mathcal{D}(\widehat{H}_{m_1}) = \mathcal{D}(\widehat{H}_{m_2})$ for every two real masses.*

*Proof.* The weighted energy series is the sum of the nonnegative momentum series and $m^2 \sum_n \|a_n\|^2$. The second series is finite. Thus the energy series is finite exactly when the momentum series is finite. Adding the already finite unweighted series gives precisely the displayed inhomogeneous first-order condition. This condition contains no mass parameter. □

The inhomogeneous weight retains the zero mode even when $m = 0$. There is no loss of the ambient square-summability requirement in the massless case. The phrase first-order Fourier domain refers to this exact condition. It agrees with the conventional Fourier definition of periodic $H^1$ regularity; this manuscript does not claim a new checked equivalence with a separate weak-derivative implementation.

### 19.1 A concrete domain boundary

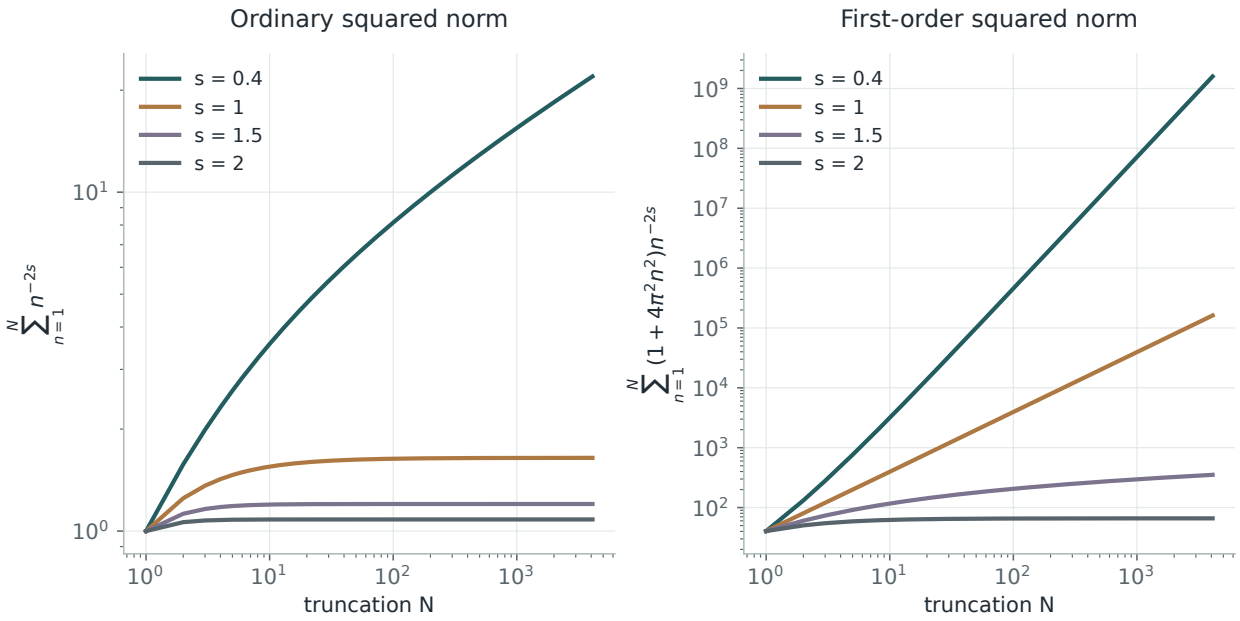
As an ordinary example, put nonzero coefficients only on the positive axis modes $n = (k, 0, 0)$, $k \geq 1$, and set $a_n = k^{-s} v$ for a fixed unit spinor $v$. All remaining coefficients are zero. Then

$$\sum_n \|a_n\|^2 = \sum_{k \geq 1} k^{-2s},$$

which is finite exactly when $s > 1/2$. The additional momentum series is

$$4\pi^2 \sum_{k \geq 1} k^{2-2s},$$

which is finite exactly when $s > 3/2$. Thus the choice $s = 1$ gives an $L^2$ field outside the Hamiltonian domain, while $s = 2$ gives one inside it. These thresholds use axis-supported data. A radially populated three-dimensional family has a different counting factor.




The norm condition and the Hamiltonian-domain condition impose different decay requirements. The exact threshold argument uses the infinite series; finite plotted partial sums only illustrate its behavior.

## 20 Exact operator energy

For $a \in \mathcal{D}(\widehat{H}_m)$, summing the modal norm identity gives

$$\left\|\widehat{H}_m a\right\|^2 = \sum_n (m^2 + |p(n)|^2) \left\|a_n\right\|^2 .$$

This is the squared norm of the Hamiltonian output. It is nonnegative. It must be distinguished from the expectation $\left\langle a, \widehat{H}_m a \right\rangle$, which may have either sign. The finite-packet discussion of positive and negative energy branches already exhibits that distinction.

An immediate ordinary consequence is the explicit graph norm

$$\|a\|^2_{\text{graph}} = \sum_n (1 + m^2 + |p(n)|^2) \left\|a_n\right\|^2 .$$

The graph norm is therefore equivalent to the norm defined by the first-order Fourier weight. Writing $w(n) = 1 + |p(n)|^2$, one has

$$w(n) \le 1 + m^2 + |p(n)|^2 \le (1 + m^2) w(n).$$

The domains coincide for all masses even though these graph norms and the operator actions depend on the mass.

## 21 The Hamiltonian on physical spinor fields

Use the checked Fourier equivalence to define

$$\mathcal{D}(H_m^{\text{phys}}) = \mathcal{F}^{-1}\mathcal{D}(\widehat{H}_m), \qquad H_m^{\text{phys}} f = \mathcal{F}^{-1}\widehat{H}_m \mathcal{F} f.$$

The second formula is used only for $f$ in the displayed domain. Its Fourier intertwining identity is

$$\mathcal{F}(H_m^{\text{phys}} f) = \widehat{H}_m(\mathcal{F} f).$$

**Theorem 21.1** (Physical torus Hamiltonian)**.** *The operator $H_m^{\text{phys}}$ is densely defined, symmetric, self-adjoint, and closed on $\mathcal{H} = L^2(\mathbb{T}^3; \mathbb{C}^4)$. Its domain consists exactly of fields satisfying*

$$\sum_{n \in \mathbb{Z}^3} (1 + |2\pi n|^2) \left\|\hat{f}(n)\right\|^2 < \infty,$$

*and for every domain vector,*

$$\left\|H_m^{\text{phys}} f\right\|^2_{\mathcal{H}} = \sum_n (m^2 + |2\pi n|^2) \left\|\hat{f}(n)\right\|^2 .$$

*Proof.* The domain characterization is the coefficient characterization applied to $\mathcal{F} f$. Density transports through the isometric equivalence. For two domain vectors, transporting the inner product and using coefficient-space symmetry proves physical-space symmetry.

To prove the reverse adjoint-domain inclusion, let $g$ have physical adjoint value $z$. For every coefficient domain vector $a$, test the physical adjoint identity at $\mathcal{F}^{-1}a$. Fourier isometry gives

$$\left\langle \widehat{H}_m a, \mathcal{F}\, g \right\rangle = \langle a, \mathcal{F}\, z \rangle \,.$$

Thus $\mathcal{F}\, g$ belongs to the coefficient adjoint domain with adjoint value $\mathcal{F}\, z$. Coefficient self-adjointness puts $\mathcal{F}\, g$ in the maximal coefficient domain, hence puts $g$ in the physical domain. The actions agree. Symmetry supplies the opposite inclusion, giving equality with the adjoint. Closedness follows, and Fourier isometry transports the exact energy sum. □

On finite Fourier polynomials, the operator agrees with the matrix differential expression $-i\sum_j \alpha_j \partial_j + m\beta$ already calculated. Its extension is specified by Fourier transport and the maximal domain, so no pointwise differentiability assumption is imposed on every $L^2$ representative. Equality with a separately constructed weak differential operator is a further interface statement if that alternate realization is introduced.

# The operator and its evolution

## 22 The completed operator construction

The free field now has two precise mathematical descriptions. Modewise exponentials define a unitary evolution on the complete coefficient space and, by Fourier conjugation, on physical torus fields. The maximal coefficient multiplier defines a self-adjoint unbounded Hamiltonian whose domain is the first-order weighted Fourier space.

These constructions answer different questions. Norm preservation determines the behavior of the bounded evolution maps. Self-adjointness determines the operator and its adjoint domain. Finite-packet graph approximation controls the state and its Hamiltonian image at the same time. No one of these statements can replace the others.

| Construction | Mathematical role |
|---|---|
| Clifford mode matrices | Exact mass shell and finite-dimensional self-adjointness. |
| Mode exponentials | Group identities and unitary spinor dynamics. |
| Completed coefficient evolution | Removal of finite support by norm stability. |
| Fourier equivalence | Lossless transport between coefficients and physical fields. |
| Maximal Hamiltonian | Density, adjoint-domain equality, and closedness. |
| Weighted domain | A mass-independent characterization of first-order regularity. |

The explicit spectral-projector formula, sharp finite Duhamel coefficient, and worked examples are deductions with proofs in the text. The technical appendix distinguishes these deductions from their exact formal correspondences. Recorded compiler evidence applies to the named statements and hypotheses, not to every sentence of interpretation.

## 23 Connecting the group to its generator

The next chapter proves that the constructed group is the spectral evolution of this Hamiltonian:

$$\mathcal{U}(t) = e^{-itH_m^{\mathrm{phys}}}.$$

The proof establishes strong continuity and differentiates on the full Hamiltonian domain with control of the Fourier tail. The reverse domain inclusion then shows that the infinitesimal generator has exactly that domain. The equality is therefore a conclusion about the complete Hilbert-space group, not an inference from agreement on finitely many modes.

The full-domain generator identification has a checked Lean implementation. The abstract spectral-functional-calculus identity remains a written deduction without a separate formal API linkage. The technical appendix distinguishes their verification coverage; no additional analytic hypothesis is introduced.

## 24 Operator limits and regularity

The centered family acts on the same infinite Fourier index set as the continuum family. The generator chapter proves convergence on the Hamiltonian domain and strong-resolvent convergence. It also gives a quadratic spacing estimate for the propagator, uniformly on bounded time intervals, under an explicit third-order spatial Fourier moment condition.

The regularity condition is essential to that estimate. Arbitrary square-integrable data retain the strong compact-time convergence statement, but are not assigned a universal quadratic rate. The first-order Hamiltonian domain alone is not substituted for the third-order moment used by the quantitative bound.

A changing finite spatial grid is a different comparison problem. It needs sampling, aliasing control, embeddings into the physical Hilbert space, and compatibility with geometric refinement. The centered multiplier family does not supply those ingredients merely by using a spacing parameter.

## 25 Beyond the free field

A prescribed background connection changes the operator and its domain. Dynamical gauge fields additionally require field energy, a Gauss constraint, matter backreaction, and an existence theory. The gauge chapter develops the finite transport and refinement ingredients for these questions without treating them as a coupled continuum evolution theorem.

The gravity, arithmetic refinement, and cosmology chapters retain their own mathematical and physical assumptions. The free model provides a controlled analytic sector to which such constructions may be compared. It neither derives a dynamical metric nor predicts an observational parameter.

## Sources and further reading


[1] Komorebi P., *Formal source for the free Dirac construction*. Exact module, compiler, and archive identities are recorded in the formal verification appendix. Its certificate applies to the stated source declarations and hypotheses.

[2] Komorebi P., *Fourier Packets and Hilbert Stability for Dirac Evolution*. The finite-packet results PH01 to PH12 provide the underlying notation. Their written deductions and exact formal correspondences are distinguished in the verification appendix.

[3] Komorebi P., *Projector Refinement Physics*: foundational and cosmological studies, technical supplement, selected second implementation, and covariant backreaction analysis. The technical appendices identify their source roles and computational evidence.

[4] P. A. M. Dirac, *The quantum theory of the electron*, Proceedings of the Royal Society of London. Series A **117** (1928), no. 778, 610–624. doi:10.1098/rspa.1928.0023.

[5] C. Roiesnel, *Definition of the covariant lattice Dirac operator*, Physical Review D **87** (2013), 074505. doi:10.1103/PhysRevD.87.074505. The symmetric difference stencil is also recorded in Eq. (9) of the author's preprint arXiv:1211.1944v1 (8 November 2012).

[6] P. Bader, S. Blanes, F. Casas, and M. Seydaoğlu, *An efficient algorithm to compute the exponential of skew-Hermitian matrices for the time integration of the Schrödinger equation*, Mathematics and Computers in Simulation **194** (2022), 383–400. doi:10.1016/j.matcom.2021.12.002. Author preprint: arXiv:2103.10132v2 (7 December 2021).